CODIMENSION ZERO POINTS OF A NORMALIZATION ARE IMAGES OF CODIMENSION ZERO POINTS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof uses the codimension zero definition through the local ring dimension of the chapter; the identification of minimal primes with codimension zero points is made explicitly.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Etale locality of the normalization and its affine description. Lemma lemma-properties-normalization, used at proof lines 10118-10120. Reduces to X = Spec(A) and identifies X' = Spec(A') with A' the integral closure in the global sections.
- Codimension zero points of an affine scheme correspond to minimal primes. Properties of Spaces, Definition spaces-properties-definition-dimension-local-ring, used at proof lines 10111-10120. Turns the point x' into a minimal prime of A'.
- Existence of an affine etale cover of Y. used at proof lines 10121-10122. Produces V = Spec(B) with B the source of an injective map from A'.
- The image of a minimal prime along an injective ring map. Algebra, Lemma algebra-lemma-injective-minimal-primes-in-image, used at proof lines 10123-10125. Shows that every minimal prime of A' comes from a minimal prime of B.

Source reference. Original lemma 00FK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 574–574. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0ABP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13027–13027. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original definition 04NA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6071–6071. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let S be a scheme. Let $f : Y \to X$ be a quasi-compact and quasi-separated morphism of schemes. Let $X' \to X$ be the normalization of X in Y. If $x' \in |X'|$ is a point of codimension 0 (Properties of Spaces, Definition 04NA), then x' is the image of some $y \in |Y|$ of codimension 0.*

Proof. By Lemma 0ABP and the definitions, we may assume that $X = \mathrm{Spec}(A)$ is affine. Then $X' = \mathrm{Spec}(A')$ where A' is the integral closure of A in $\Gamma(Y, \mathcal{O}_Y)$ and x' corresponds to a minimal prime of A'. Choose a surjective étale morphism $V \to Y$ where $V = \mathrm{Spec}(B)$ is affine. Then $A' \to B$ is injective, hence every minimal prime of A' is the image of a minimal prime of B, see Algebra, Lemma 00FK. The lemma follows. □